

Python – Seaborn

Introduction to Seaborn

What is Seaborn?

Seaborn is a Python data visualization library built **on top of Matplotlib**.

It provides a **high-level interface** for drawing attractive and informative statistical graphics with much less code.

Seaborn is especially popular in **data science and analytics** because it works seamlessly with **Pandas DataFrames** and automatically applies clean visual styles.

Why Use Seaborn?

While Matplotlib gives full control, it is **low-level** and requires more manual work. Seaborn simplifies this process by offering smart defaults and advanced plot types.

Advantages of Seaborn:

- Less code for complex plots
- Beautiful default themes and color palettes
- Direct support for Pandas DataFrames
- Built-in statistical visualizations
- Automatic handling of categorical data

In Short:

- **Matplotlib** → fine control and customization
- **Seaborn** → fast, clean, professional plots

Installing and Importing Seaborn

Installation:

```
pip install seaborn
```

Importing:

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

Seaborn uses Matplotlib internally, so `matplotlib.pyplot` is often imported alongside it.

Seaborn Themes

Seaborn allows you to control the **overall appearance** of plots using themes.

Setting a Theme:

```
sns.set_theme(style="darkgrid")
```

Common Theme Options:

- `whitegrid`
- `darkgrid`
- `white`
- `dark`
- `ticks`

Themes affect:

- Background
- Grid lines
- Font sizes
- Overall aesthetics

Simple Line Plot Using Seaborn

Example:

```
import numpy as np
```

```
x = np.arange(0, 10)
```

```
y = x ** 2
```

```
sns.lineplot(x=x, y=y)
```

```
plt.title("Simple Line Plot Using Seaborn")
```

```
plt.show()
```

Why This Looks Better Than Matplotlib:

- Automatic smoothing
- Clean grid
- Better default colors

Built-in Datasets in Seaborn

Seaborn provides **ready-to-use datasets** for practice and learning.

These datasets simulate **real-world data**, including:

- Numerical values
- Categorical variables
- Missing data

Viewing Available Datasets

```
print(sns.get_dataset_names())
```

Examples of Common Datasets:

- tips
- iris
- penguins
- titanic
- diamonds
- flights

Loading a Seaborn Dataset

```
data = sns.load_dataset("penguins")
```

```
print(data.head())
```

This returns a **Pandas DataFrame**, which can be used directly for plotting.

Why Seaborn Provides Datasets

- To learn plotting without finding data
- To practice real-world scenarios
- To demonstrate statistical plots easily
- To save time during learning and teaching

Basic Plot Types in Seaborn

1. Line Plot

Used to show trends between two numeric variables.

Example:

```
sns.lineplot(x="bill_length_mm", y="bill_depth_mm", data=data)
plt.title("Line Plot Example")
plt.show()
```

2. Scatter Plot

Used to show relationships between two variables and optionally categorize data using color.

Example:

```
sns.scatterplot(
    x="bill_length_mm",
    y="bill_depth_mm",
    hue="species",
    data=data
)
plt.title("Scatter Plot with Hue")
plt.show()
```

Advantage:

- Automatically creates legend
- Color-coding made easy with hue

3. Bar Plot

Bar plots in Seaborn show **aggregated values** (mean by default).

Example:

```
sns.barplot(x="species", y="body_mass_g", data=data)

plt.title("Average Body Mass by Species")

plt.show()
```

Key Difference from Matplotlib:

- Seaborn computes statistics automatically

4. Box Plot

Box plots show:

- Median
- Quartiles
- Spread
- Outliers

Example:

```
sns.boxplot(x="species", y="body_mass_g", data=data)

plt.title("Body Mass Distribution by Species")

plt.show()
```

Use Case:

Great for comparing distributions across categories.

5. Heatmap

Heatmaps are used to visualize **matrix-like data**, often correlations.

Example:

```
import pandas as pd
```

```
scores = pd.DataFrame({
```

```
"Math": [80, 70, 90, 85],  
"Science": [75, 88, 92, 80],  
"English": [78, 85, 88, 90]  
})
```

```
sns.heatmap(scores.corr(), annot=True, cmap="coolwarm")  
  
plt.title("Correlation Heatmap")  
  
plt.show()
```

Working with Pandas DataFrames in Seaborn

One of Seaborn's biggest strengths is **direct DataFrame support**.

Example:

```
import pandas as pd  
  
df = pd.DataFrame({  
    "experience": [1, 2, 3, 4, 5, 6],  
    "salary": [25000, 30000, 40000, 52000, 65000, 75000],  
    "role": ["Junior", "Junior", "Mid", "Mid", "Senior", "Senior"]  
})  
  
sns.scatterplot(  
    x="experience",  
    y="salary",  
    hue="role",  
    data=df  
)  
  
plt.title("Salary vs Experience")  
  
plt.show()
```